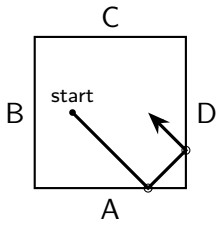
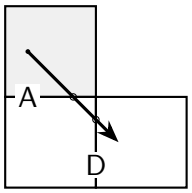
A shot starts inside the room and bounces off its walls. Record the wall letters in the order hit. Stop after the last letter you want to record. A shot that hits a corner is not allowed.

One partner draws a shot with a ruler; the other checks it. To check a bounce, reflect the next room across the wall. The shot continues in a straight line through the reflected copy. Swap jobs for the next shot.

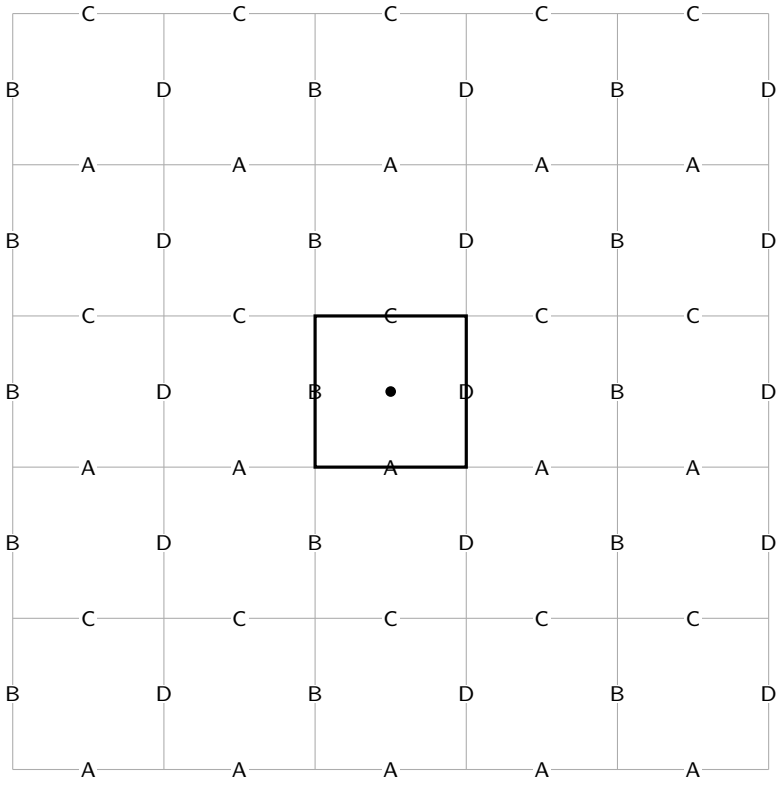


first A, then D: AD

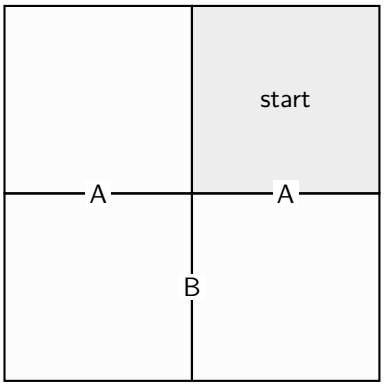


the same shot through reflected copies

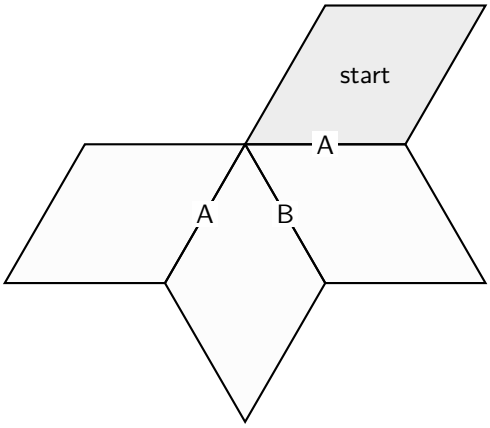
Problem 1: Find six different three-letter bounce words in the square. Draw straight shots from the dot through the reflected copies. Write each word by the end of its shot.



Problem 2: Can a shot begin ABA in each room? Start anywhere inside the shaded room. Try to cross A, then B, then A with one straight line, without leaving the drawn copies. Draw a shot wherever it works. Explain any impossibility.

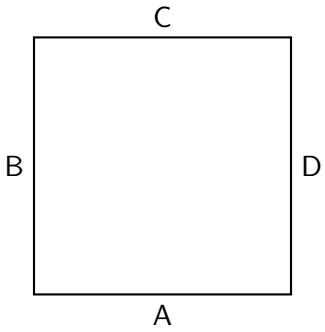


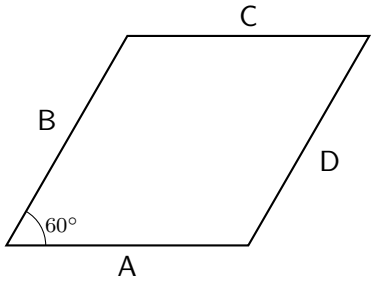
square copies



$60°$ rhombus copies

Copy a successful shot into its single room. Use a mirror on each wall to check that the incoming line and the reflection of the outgoing line line up.

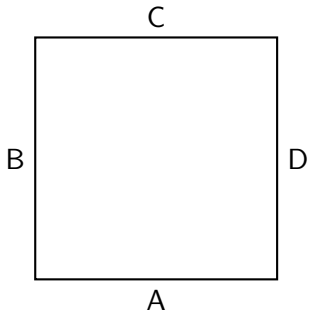




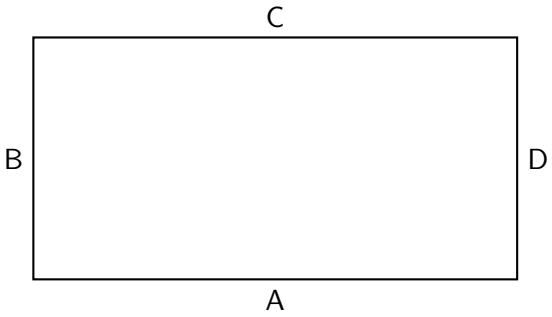
Problem 3: One partner secretly chooses a room and draws a checked shot with at least three bounces. Give only its word. The other partner circles the rooms they have not ruled out. To rule out a room, explain why the word is impossible there. Then reveal the shot and swap jobs.

You may reuse a checked shot from Problems 1–2. For new shots, use the square and rectangle windows on page 4, or the mirror and large rhombus rooms on page 5.

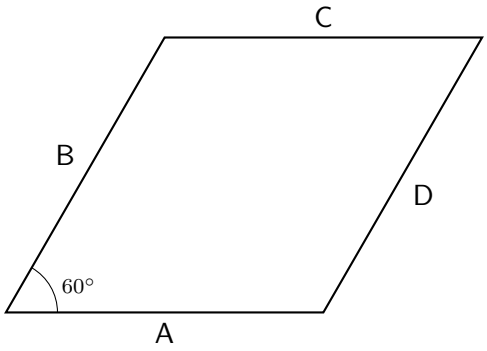
Failing to find a shot is not a reason to rule out a room.



square



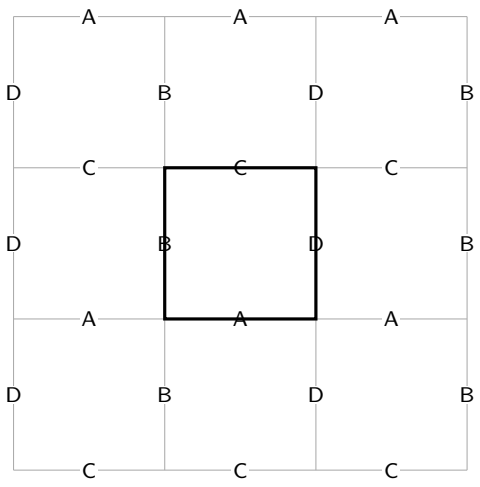
wide rectangle

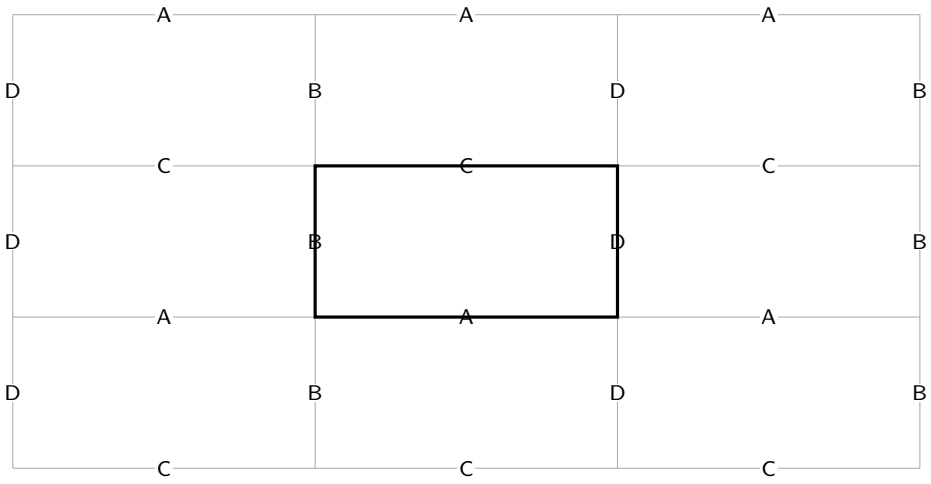


60° rhombus

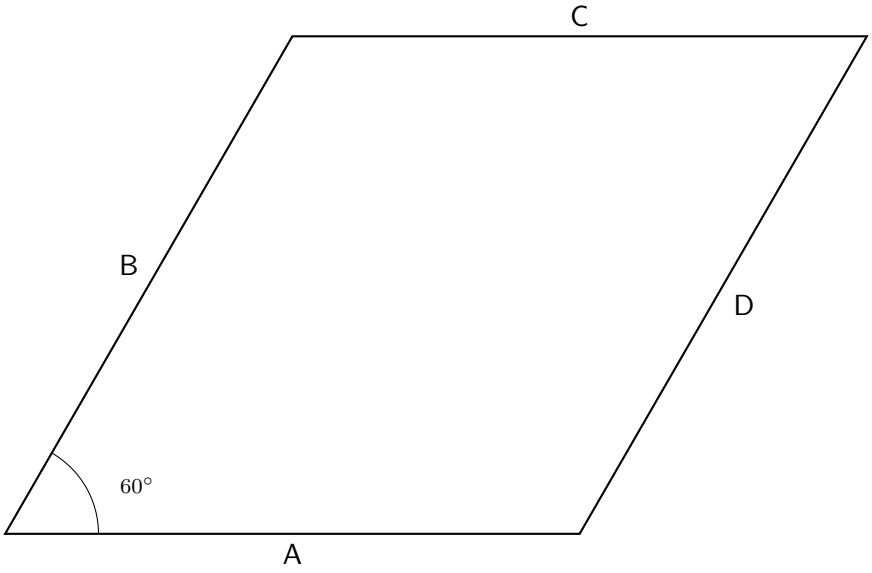
word	rooms not ruled out		
_____	square	wide rectangle	rhombus
_____	square	wide rectangle	rhombus
_____	square	wide rectangle	rhombus

Problem 4: Draw three shots starting inside the bold square, with different words of at least three bounces. Each shot must hit both a horizontal and a vertical wall. Move each shot to the wide rectangle without changing its bounce word. Could any bounce word tell these two rooms apart? Explain why, or find a word that only one room can make.

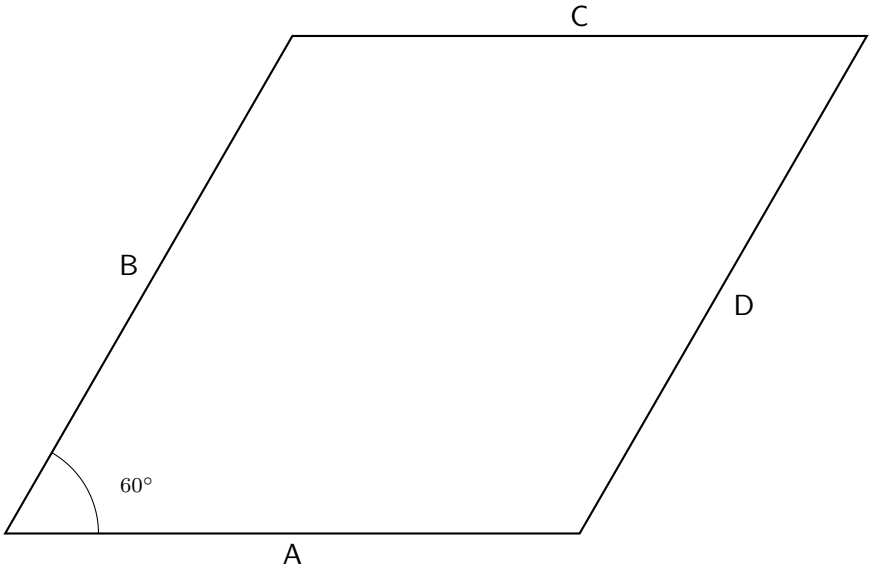




Extra workspace for Problems 2–3.



word: _____



word: _____